

**Absolute Value and Exponential Functions Quiz 1****DIRECTIONS**

10 items • 12 points • 25 minutes • Form A

Choose one answer unless an item says Select all. Show work inside each item panel. Label graphs, endpoints, units, and constraints. Use a calculator only where the item says CALCULATOR AVAILABLE. No reference sheet is provided.

1

SOLVE • COORDINATE GRAPH • CALCULATOR AVAILABLE

1 PT

State the range of $f(x) = 2|x + 5| + 1$.

- ☐ A $y \leq 1$
- ☐ B $y \geq -5$
- ☐ C $y \geq 1$
- ☐ D all real numbers

2

SOLVE • SYMBOLIC • CALCULATOR AVAILABLE

1 PT

Given $f(x) = -|x - 5| + 5$, find $f(8)$.

3

SOLVE • COORDINATE GRAPH • NO CALCULATOR

1 PT

State the range of $f(x) = -|x - 1| + 5$.

- ☐ A all real numbers
- ☐ B $y \geq 5$
- ☐ C $y \geq 1$
- ☐ D $y \leq 5$

4

CONNECT • COORDINATE GRAPH • NO CALCULATOR

1 PT

State the vertex and opening direction of $f(x) = 2|x - 4| + 2$. Select all statements that are true.

- ☐ A The result is not vertex (0, 2); opens up.
- ☐ B The result is vertex (4, -2); opens down.
- ☐ C The result is vertex (4, 2); opens up.
- ☐ D The result is vertex (-4, 2); opens up.

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CONNECT • MULTIPLE REPRESENTATION • NO CALCULATOR

1 PT

Match each function to its vertex. Each one uses key features.

1.	An absolute-value function has $f(0) = 4$, $f(1) = 5$, and $f(2) = 4$. Give the vertex.	Match: _____
2.	Given $f(x) = 2 x + 4 + 1$, find $f(-2)$.	Match: _____
3.	State the vertex and opening direction of $f(x) = 2 x + 1 - 3$.	Match: _____

Answer bank: A. (1, 5) • B. 5 • C. vertex (-1, -3); opens up

6

SOLVE • SYMBOLIC • CALCULATOR AVAILABLE

1 PT

Factor $x^2 - 9$.

7

SOLVE • COORDINATE GRAPH • CALCULATOR AVAILABLE

1 PT

State the vertex and opening direction of $f(x) = 2|x + 2| + 1$.

- ☐ A vertex (-2, 1); opens up
- ☐ B vertex (0, 1); opens up
- ☐ C vertex (-2, -1); opens down
- ☐ D vertex (2, 1); opens up

8

APPLY + EXPLAIN • TABLE • CALCULATOR AVAILABLE

2 PTS

An absolute-value function has $f(-2) = -2$, $f(-1) = -4$, and $f(0) = -2$. Give the vertex. Show enough reasoning to justify your answer.

9

CHECK + JUSTIFY • TABLE • NO CALCULATOR

1 PT

A quadratic has $f(-6) = 8$, $f(-5) = 7$, and $f(-4) = 8$. Give the vertex. A student reported (-6, 8). Identify the likely error and give the correct result.

10

APPLY + EXPLAIN • COORDINATE GRAPH • NO CALCULATOR

2 PTS

Apply the lesson strategy to this unfamiliar version. State the range of $f(x) = 2|x| - 2$. State the result and justify why it is reasonable.
